Testing evidence:

Test environment:

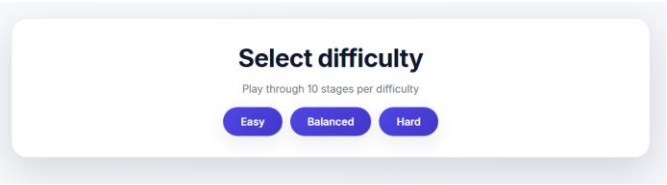
Windows 11, python 3.14.3, node v25.6.0, Chrome browser

backend running on localhost:5000 and client on localhost:5173

Manual end to end tests:

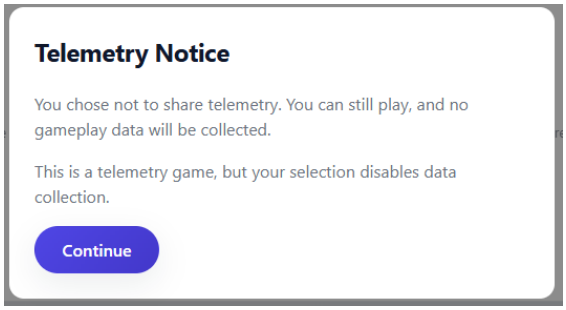
Scenario 1:

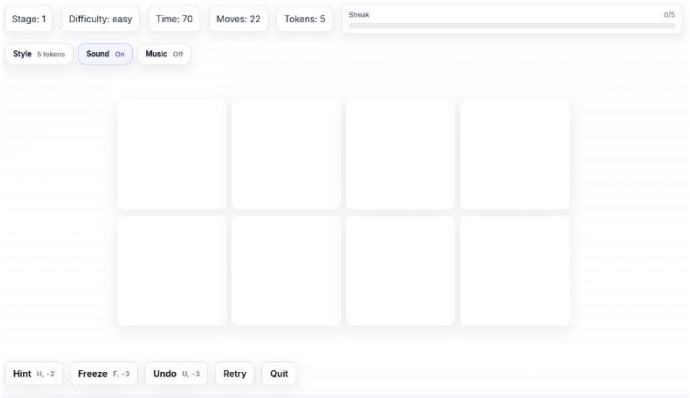
- Expected Results
  - Consent is stored, modal doesn't reappear on refresh
  - Difficulty screen loads
- Actual Restult
  - Consent was stored, modal didn't reappear on refresh
  - When selecting to be a player, difficulty screen loads and appears
- Screenshot(s)



Scenario 2:

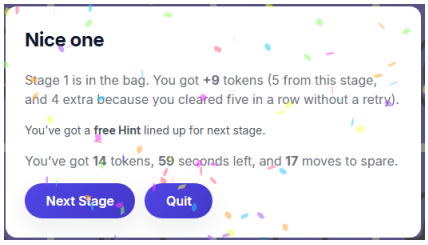
- Expected Results
  - Game remains to be playable after declining
  - No crashes
- Actual Results
  - Shown decline notice
  - Game remains playable
  - No crashes
- Screenshot(s)





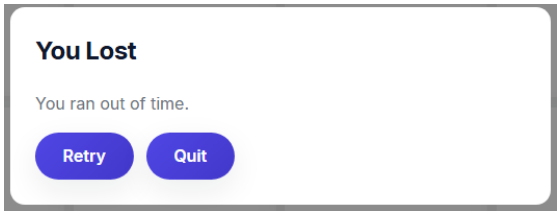
Scenario 3:

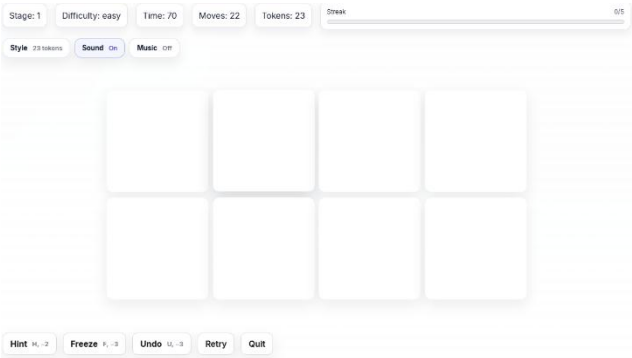
- Expected results
  - Win modal is displayed with each metric
  - Advancing does not error
- Actual results
  - Win modal is displayed
  - Advancing does not error
- Sreenshot(s)



Scenario 4:

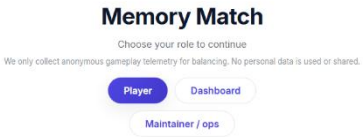
- Expected results
  - Fail modal is shown
  - Board and metrics reset after “retry” is clicked
- Actual results
  - Fail model is shown with reason of failure
  - Board and metrics get reset after retry
- Screenshot(s)

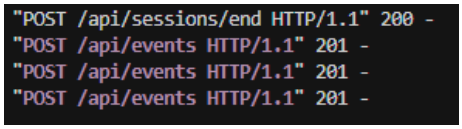




Scenario 5:

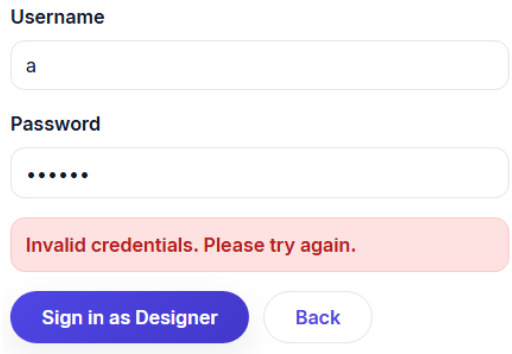
- Expected results
  - When quit, returns to main menu
  - No error in console
- Actual results
  - Quitting returns to main menu
  - No error in console
- Screenshot(s)





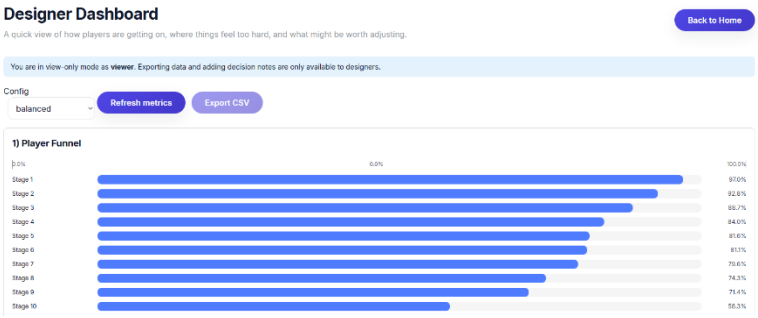
Scenario 6:

- Expected results
  - Error message is shown
  - Not taken to dashboard
- Actual results
  - Error message shown
  - Not taken past the sign in page
- Screenshot(s)



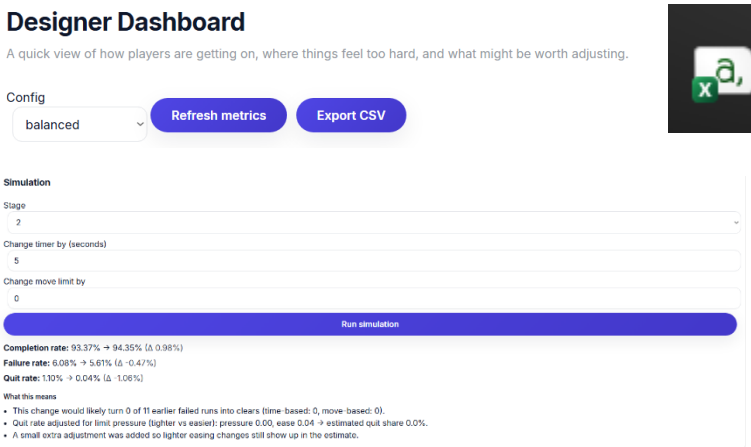
Scenario 7:

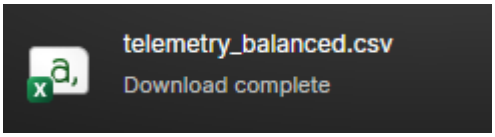
- Expected results
  - Dashboard loads data to be viewed
  - Export and write actions are disabled
- Actual results
  - Dashboard loads
  - Export and write actions are disabled
- Screenshot(s)



Scenario 8:

- Expected results
  - Dashboard loads data to be viewed, exported, and write features usable
  - CSV file downloads correctly
  - Simulation and suggestions are shown and run
- Actual results
  - Dashboard loads data in correct ways
  - CSV file downloads
  - Simulation and suggestions are shown and run
- Screenshot(s)





Automated tests:

python -m pytest:

```
collected 16 items

tests\test_balancing.py ..                                              [ 12%]
tests\test_events.py ........                                           [ 62%]
tests\test_ingestion.py ..                                              [ 75%]
tests\test_metrics.py ....                                              [100%]

============================================================ 16 passed in 10.11s ============================================================
```